PRODUCING A SMOOTH CHART THROUGH A VERSAL FINITE-TYPE FAMILY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2407-2429 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 2431-2454, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 36 of queue_artin.json (pb=1090, stmt 2407-2429, proof 2431-2454).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07XS: sources/stacks-artin/pinned-original-source.tex, source lines 2268–2275. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06IW: sources/stacks-artin/pinned-original-source.tex, source lines 2454–2454. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XH: sources/stacks-artin/pinned-original-source.tex, source lines 2092–2102. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07WU: sources/stacks-artin/pinned-original-source.tex, source lines 824–833. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XX: sources/stacks-artin/pinned-original-source.tex, source lines 2345–2354. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XT: sources/stacks-artin/pinned-original-source.tex, source lines 2281–2288. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07WV: sources/stacks-artin/pinned-original-source.tex, source lines 866–885. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XJ: sources/stacks-artin/pinned-original-source.tex, source lines 2462–2489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. Let k be a finite type field over S and let x_0 be an object of $\mathcal{X}$ over* $\mathrm{Spec}(k)$ *with image $s \in S$. Assume*

1. *$\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces,*
2. *$\mathcal{X}$ satisfies axioms [1], [2], [3] (see Section 07XJ),*
3. *every formal object of $\mathcal{X}$ is effective,*
4. *openness of versality holds for $\mathcal{X}$, and*
5. *$\mathcal{O}_{S,s}$ is a G-ring.*

Then there exist a morphism of finite type $U \to S$ and an object x of $\mathcal{X}$ over U such that

$$x : (Sch/U)_{fppf} \longrightarrow \mathcal{X}$$

is smooth and such that there exists a finite type point $u_0 \in U$ whose residue field is k and such that $x|_{u_0} \cong x_0$.

Proof. By axiom [2], Lemma 07WU, and Remark 07WV we see that $\mathcal{F}_{\mathcal{X},k,x_0}$ satisfies (S1) and (S2). Since also the tangent space has finite dimension by axiom [3] we deduce from Formal Deformation Theory, Lemma 06IW that $\mathcal{F}_{\mathcal{X},k,x_0}$ has a versal formal object ξ. Assumption (3) says ξ is effective. By axiom [1] and Lemma 07XH there exists a morphism of finite type $U \to S$, an object x of $\mathcal{X}$ over U, and a finite type point u_0 of U with residue field k such that x is versal at u_0 and such that $x|_{\mathrm{Spec}(k)} \cong x_0$. By openness of versality we may shrink U and assume that x is versal at every finite type point of U. We claim that

$$x : (Sch/U)_{fppf} \longrightarrow \mathcal{X}$$

is smooth which proves the lemma. Namely, by Lemma 07XT x satisfies (07XS) whereupon Lemma 07XX finishes the proof. □